

Analysis of Resonant and Non-Resonant Wireless Power Transfer Systems for Biomedical Implants

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Abstract—This work investigates resonant and non-resonant inductively coupled wireless power transfer (WPT) systems operating at 5 MHz for biomedical implant applications. Two planar spiral coil families were analyzed using LTSpice simulations for both 2-coil and 3-coil topologies. Frequency sweeps from 2 MHz to 8 MHz and a parametric distance sweep from 0 mm to 40 mm were performed to evaluate power transfer efficiency (PTE). The results demonstrate that resonant tuning significantly improves transfer efficiency compared to non-resonant operation. In addition, the 3-coil resonant topology exhibits improved tolerance to coil separation due to enhanced magnetic coupling through the relay coil. Experimental validation of the resonant 2-coil system was carried out using a Vector Network Analyzer (VNA). Measured S_{21} values decreased from -18.75 dB at 10 mm to -31.03 dB at 40 mm, corresponding to an efficiency reduction from approximately 1.33% to 0.079%. Although the measured efficiencies were lower than simulated values, the experimental results confirmed the expected distance-dependent transfer behavior. The difference between simulation and measurement is attributed to practical effects such as coil losses, parasitic elements, alignment error, and impedance mismatch.

I. INTRODUCTION

Wireless power transfer (WPT) has gained significant attention for biomedical implant applications because it enables power delivery without batteries or transcutaneous wires. Conventional wired powering methods increase the risk of infection, reduce patient comfort, and limit long-term implant reliability. Inductive coupling is widely used for implantable systems due to its relatively simple implementation and safe near-field operation. However, the efficiency of inductive power transfer decreases rapidly as the separation between coils increases.

Resonant inductive coupling improves transfer efficiency by maximizing energy exchange near the operating frequency. In addition, multi-coil topologies can improve magnetic coupling and reduce the dependence on direct transmitter–receiver interaction. In particular, a relay coil in a 3-coil system can enhance power transfer over larger distances compared to a conventional 2-coil link.

In this work, resonant and non-resonant 2-coil and 3-coil WPT systems operating at 5 MHz are analyzed using LTSpice simulations. Frequency response and distance-dependent power transfer efficiency are evaluated for different configurations. Experimental validation of the resonant 2-coil system is also performed using Vector Network Analyzer (VNA) measurements to compare simulated and practical performance.

II. METHODS

A. Coil Design

Two planar spiral coil families were used in this work. Coil A was designed as the smaller, higher-inductance coil, while Coil B was designed as the larger, lower-inductance coil. The coils were fabricated using 20 AWG copper wire with a turn spacing of 1 mm. Both source and load resistances were fixed at 50Ω .

For the resonant configurations, series capacitors were selected such that the resonant tanks were tuned to an operating frequency of 5 MHz. In the 3-coil topology, the intermediate relay coil was placed between the transmitter and receiver coils to improve magnetic coupling and enhance power transfer efficiency.

B. Simulation Setup

LTSpice AC analysis was used to evaluate both resonant and non-resonant 2-coil and 3-coil WPT systems. Frequency sweeps were performed from 2 MHz to 8 MHz to analyze the transfer characteristics around the resonant frequency. The coupling coefficients used in the simulations were $k = 0.01$ for the source-side link and $k = 0.3$ for the relay-to-load link.

Power transfer efficiency (PTE) was calculated by comparing the power delivered to the load with the transmitted input power. Resonant cases were simulated using tuning capacitors, while non-resonant cases were analyzed without resonant compensation.

A separate parametric sweep was performed by varying the coil separation distance from 0 mm to 40 mm in order to study the effect of distance on transfer efficiency for the resonant 2-coil and 3-coil systems.

C. Experimental Validation

Experimental validation was carried out for the resonant 2-coil WPT system using a Vector Network Analyzer (VNA).

The transmission parameter S_{21} was measured at separation distances of 10 mm, 20 mm, 30 mm, and 40 mm. The measured S_{21} values were converted to approximate efficiency using

$$\eta \approx 10^{(S_{21}/10)} \times 100\% \quad (1)$$

The measured S_{21} values were -18.75 dB, -23.6 dB, -28.4 dB, and -31.03 dB, corresponding to efficiencies of approximately 1.33%, 0.44%, 0.14%, and 0.079%, respectively.



Fig. 1. Experimental setup used for VNA-based validation of the resonant 2-coil wireless power transfer system.

III. RESULTS

A. Resonant vs. Non-Resonant 2-Coil WPT System

The power transfer efficiency (PTE) of the resonant and non-resonant 2-coil WPT systems was evaluated over a frequency range of 2 MHz to 8 MHz. The resonant 2-coil system exhibited a clear efficiency peak near the designed operating frequency of 5 MHz, while the non-resonant configuration showed significantly lower PTE across the frequency range.

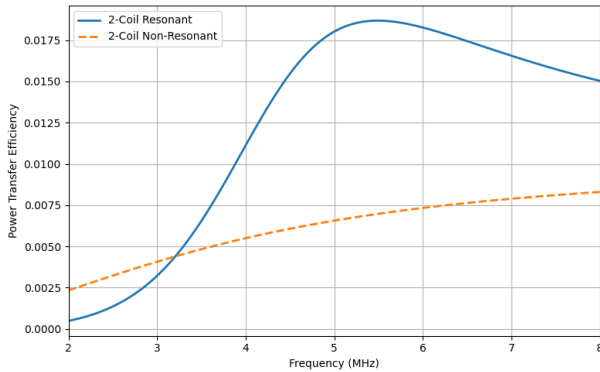


Fig. 2. Power transfer efficiency versus frequency for resonant and non-resonant 2-coil WPT systems.

The results demonstrate that resonant tuning is essential for efficient inductive power transfer. At resonance, energy transfer between the transmitter and receiver coils is maximized

due to improved impedance matching and stronger magnetic coupling.

B. Resonant vs. Non-Resonant 3-Coil WPT System

The resonant and non-resonant 3-coil WPT systems were analyzed using frequency sweeps from 2 MHz to 8 MHz. The resonant 3-coil topology achieved higher PTE near the operating frequency of 5 MHz compared to the non-resonant configuration.

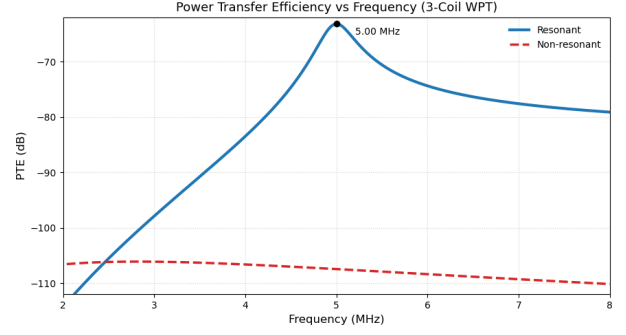


Fig. 3. Power transfer efficiency versus frequency for resonant and non-resonant 3-coil WPT systems.

The addition of the relay coil improved coupling between the transmitter and load-side coils, resulting in enhanced power transfer characteristics. Similar to the 2-coil system, the non-resonant 3-coil configuration remained inefficient across the frequency sweep, highlighting the importance of resonant operation.

C. Parametric Distance Study

A parametric analysis was performed to study the variation of PTE with respect to the separation distance between coils. The distance was varied from 0 mm to 40 mm for both resonant 2-coil and resonant 3-coil systems.

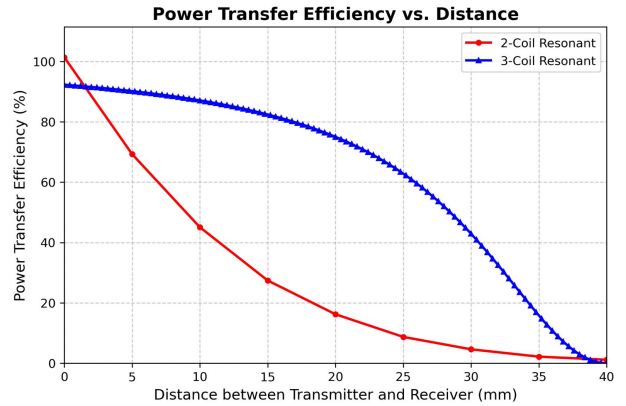


Fig. 4. Variation of power transfer efficiency with coil separation distance for resonant 2-coil and resonant 3-coil WPT systems.

As shown in Fig. 4, the PTE decreases with increasing distance for both topologies due to reduced magnetic coupling.

D. Experimental Validation

Experimental validation of the resonant 2-coil WPT system was performed using a Vector Network Analyzer (VNA). The transmission coefficient S_{21} was measured at coil separations of 10 mm, 20 mm, 30 mm, and 40 mm, and converted to power transfer efficiency (PTE).

The measured S_{21} values were -18.75 dB, -23.6 dB, -28.4 dB, and -31.03 dB, corresponding to efficiencies of approximately 1.33%, 0.44%, 0.14%, and 0.079%, respectively.

A comparison with the simulated results from the parametric distance study (Section III-C) shows that the experimental and simulated results follow the same decreasing trend with increasing coil separation. However, the magnitude of PTE differs significantly. For example, at a separation of 10 mm, the simulated PTE is on the order of tens of percent (approximately 40–50%), whereas the measured efficiency is only 1.33%. Similarly, at 20 mm and 30 mm, simulation predicts efficiencies in the range of 15–20% and 4–6%, respectively, while the measured values drop to 0.44% and 0.14%.

This discrepancy between simulation and experiment arises due to several non-ideal effects that are not fully captured in LTSpice simulations. These include resistive losses in the copper coils, skin and proximity effects at 5 MHz, parasitic capacitances between turns, and imperfect impedance matching between the source and load. Additionally, the simulation assumes ideal alignment and a constant coupling coefficient, whereas in practice, slight angular or positional misalignment between coils significantly reduces mutual inductance and hence S_{21} .

Measurement-related losses also contribute to reduced efficiency, including cable losses, connector losses, and calibration limitations of the VNA. These effects collectively lead to lower experimentally observed power transfer efficiency.

Despite the quantitative differences, the experimental results validate the key trends predicted by simulation. The monotonic decrease of PTE with increasing distance confirms the behavior observed in Section III-C, while the presence of maximum transfer near the operating frequency (5 MHz) is consistent with the resonance behavior observed in Section III-A.

Overall, the experimental results confirm the validity of the simulation model in predicting system behavior, while also highlighting the impact of practical implementation losses in real-world wireless power transfer systems.

TABLE I
MEASURED EXPERIMENTAL RESULTS FOR RESONANT 2-COIL WPT SYSTEM

Distance (mm)	S_{21} (dB)	Efficiency (%)
10	-18.75	1.33
20	-23.6	0.44
30	-28.4	0.14
40	-31.03	0.079

IV. DISCUSSION

The results obtained from both simulation and experimental validation demonstrate the importance of resonant tuning in

wireless power transfer systems. In both the 2-coil and 3-coil configurations, the resonant systems achieved significantly higher power transfer efficiency compared to the non-resonant systems. This improvement occurs because resonance enhances energy exchange between coupled coils and improves impedance matching near the operating frequency of 5 MHz.

The 3-coil resonant topology exhibited better performance over increasing separation distances compared to the 2-coil resonant system. The relay coil acts as an intermediate coupling stage, reducing the dependence on direct transmitter-to-receiver magnetic coupling. As a result, the 3-coil system maintained higher PTE across the distance sweep and showed improved tolerance to coil separation. This characteristic is especially important in biomedical implant applications, where the spacing and alignment between external and implanted coils cannot always be precisely controlled.

The experimental VNA measurements followed the same overall trend observed in simulation, where transfer efficiency decreased with increasing coil separation. However, the measured efficiencies were significantly lower than the simulated results. Several practical factors contribute to this difference. Real coils experience ohmic losses, skin effect, and parasitic capacitances that are not fully captured in ideal simulations. In addition, impedance mismatch, cable losses, and measurement setup imperfections reduce the measured transfer efficiency.

Another important source of error is coil alignment. The simulations assume ideal positioning and constant coupling coefficients, whereas the experimental setup is affected by angular misalignment and positional variation between the coils. These effects reduce effective magnetic coupling and lower the measured S_{21} values.

One limitation of this work is that experimental validation was performed only for the resonant 2-coil WPT system. Experimental analysis of the 3-coil resonant system would provide a more complete comparison between the two topologies. In addition, the parametric study considered only separation distance, while practical WPT systems are also sensitive to lateral displacement and orientation changes. Despite these limitations, the results clearly demonstrate that resonant multi-coil WPT systems provide improved transfer performance for biomedical implant applications.

V. CONCLUSION

This work analyzed resonant and non-resonant inductively coupled wireless power transfer systems for biomedical implant applications using both simulation and experimental validation. Frequency sweep analysis showed that resonant tuning significantly improves power transfer efficiency compared to non-resonant operation. The resonant 3-coil topology demonstrated better tolerance to coil separation and maintained higher efficiency over distance than the resonant 2-coil system due to improved magnetic coupling through the relay coil.

Experimental validation using VNA measurements confirmed the expected decrease in transfer efficiency with increasing separation distance. Although the measured efficien-

cies were lower than simulated values because of practical losses and non-ideal effects, the experimental results followed the same overall trend as the simulations.

Overall, the results demonstrate the effectiveness of resonant multi-coil WPT systems for biomedical implant power delivery and highlight the importance of resonance, coil coupling, and system topology in improving wireless power transfer performance.

ACKNOWLEDGMENT

Tejaswi developed the LTSpice models for the resonant and non-resonant 2-coil WPT systems. Satyam developed the LTSpice model for the non-resonant 3-coil WPT system and fabricated the experimental coils. Anirudh developed the resonant 3-coil LTSpice model, implemented the Python script used for the parametric distance study, and contributed to report preparation along with Tejaswi. All team members participated in the VNA-based experimental validation and data collection.

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